

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

22

Q1

ALGEBRA

A teacher is creating an assignment worth 70 points. The assignment will consist of questions worth 1 point and questions worth 3 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 3-point questions?

- A. $4xy = 70$
- B. $4x + y = 70$
- C. $3x + y = 70$
- D. $x + 3y = 70$

$$16x + 30 = 190$$

Which equation has the same solution as the given equation?

- A. $16x = 30$
- B. $16x = 130$
- C. $16x = 160$
- D. $16x = 190$

Marisol drove 3 hours from City A to City B. The equation below estimates the distance d , in miles, Marisol traveled after driving for t hours.

$$d = 45t$$

Which of the following does 45 represent in the equation?

- A. Marisol took 45 trips from City A to City B.
- B. The distance between City A and City B is 45 miles.
- C. Marisol drove at an average speed of about 45 miles per hour.
- D. It took Marisol 45 hours to drive from City A to City B.

Q4

GRID-IN

ALGEBRA

The y -intercept of the graph of $y = -6x - 32$ in the xy -plane is $0, y$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$72x - 3 = 63$$

Which equation has the same solution as the given equation?

A. $2x - 3 = 9$

B. $2x - 3 = 56$

C. $2x - 21 = 63$

D. $2x - 21 = 70$

A veterinarian recommends that each day a certain rabbit should eat 25 calories per pound of the rabbit's weight, plus an additional 11 calories. Which equation represents this situation, where c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit's weight is x pounds?

- A. $c = 25x$
- B. $c = 36x$
- C. $c = 11x + 25$
- D. $c = 25x + 11$

Q7

GRID-IN ALGEBRA

The equation $46 = 2x + 2y$ gives the perimeter of a rectangular rug that has length x , in feet, and width y , in feet. The width of the rug is 8 feet. What is the length, in feet, of the rug?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Cathy has n CDs. Gerry has 3 more than twice the number of CDs that Cathy has. In terms of n , how many CDs does Gerry have?

A. $3n - 2$

B. $3n + 2$

C. $2n - 3$

D. $2n + 3$

The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. Which of the following is the best interpretation of 9 in this context?

- A. Tavon will keep the vine plant for 9 months.
- B. The vine plant is expected to grow 9 inches each month.
- C. The vine plant is expected to grow to a maximum length of 9 inches.
- D. The estimated length of the vine plant was 9 inches when Tavon purchased it.

Line r in the xy -plane has a slope of 4 and passes through the point $0, 6$. Which equation defines line r ?

A. $y = -6x + 4$

B. $y = 6x + 4$

C. $y = 4x - 6$

D. $y = 4x + 6$